Topic: Predator-prey systems

Question: Is the system cooperative, competitive, or predator-prey?

$$\frac{dx}{dt} = 0.14x - 0.2xy$$

$$\frac{dy}{dt} = -0.09y + 0.10y^2 + 0.12xy$$

Answer choices:

A Cooperative

B Competitive

C Predator-prey

D None of these

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Solution: C

The interaction term in the dx/dt equation is $-0.2xy$, and the interaction term in the dy/dt equation is $0.12xy$.

When these interaction terms have opposite signs, it means the system is predator-prey. In this case, we know the size of population x is decreasing because the sign of $-0.2xy$ is negative, while the size of population y is increasing because the sign of $0.12xy$ is positive.

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Topic: Predator-prey systems

Question: Is the system cooperative, competitive, or predator-prey?

$$\frac{dx}{dt} = 0.8x + 0.15xy$$

$$\frac{dy}{dt} = 0.3y + 0.16xy$$

Answer choices:

A Cooperative

B Competitive

C Predator-prey

D None of these

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Solution: A

The interaction term in the dx/dt equation is $0.15xy$, and the interaction term in the dy/dt equation is $0.16xy$.

When these interaction terms have positive signs, it means the system is cooperative. In this case, we know the size of population x is increasing because the sign of $0.15xy$ is positive, and the size of population y is also increasing because the sign of $0.16xy$ is positive.

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Topic: Predator-prey systems

Question: Is the system cooperative, competitive, or predator-prey?

$$\frac{dx}{dt} = -0.3x - 0.21xy$$

$$\frac{dy}{dt} = 0.9y - 0.36xy$$

Answer choices:

A Cooperative

B Competitive

C Predator-prey

D None of these

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Solution: B

The interaction term in the dx/dt equation is $-0.21xy$, and the interaction term in the dy/dt equation is $-0.36xy$.

When these interaction terms have negative signs, it means the system is competitive. In this case, we know the size of population x is decreasing because the sign of $-0.21xy$ is negative, and the size of population y is also decreasing because the sign of $-0.36xy$ is negative.

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